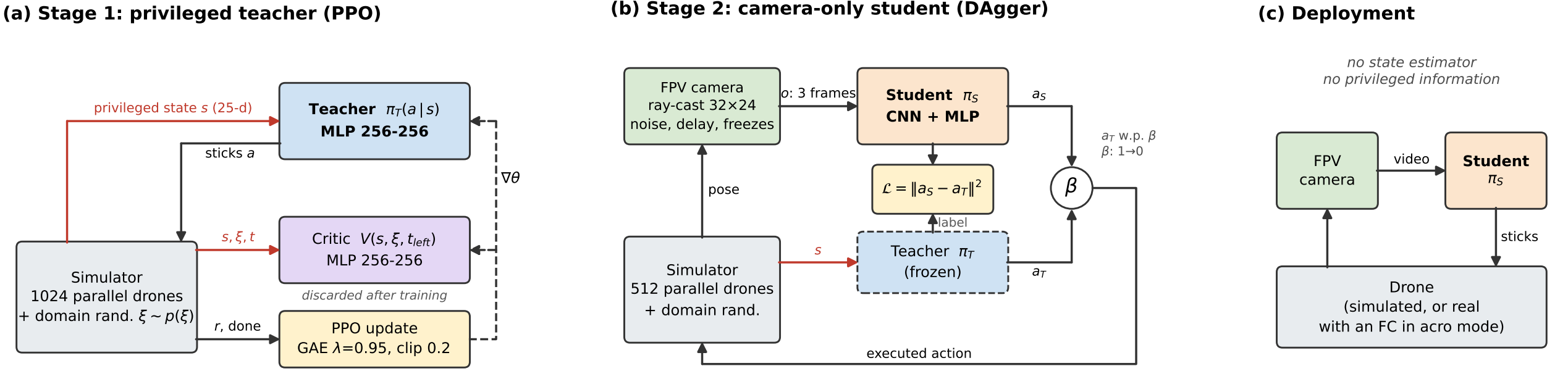
(a) Stage 1: privileged teacher (PPO)
privileged state s (25-d)
Teacher $\pi_T(a \mid s)$
MLP 256-256
sticks a
$\nabla\theta$
s, ξ, t
Critic $V(s, \xi, t_{left})$
MLP 256-256
discarded after training
Simulator
1024 parallel drones
+ domain rand. $\xi \sim p(\xi)$
r, done
PPO update
GAE λ=0.95, clip 0.2
(b) Stage 2: camera-only student (DAgger)
FPV camera
ray-cast 32×24
noise, delay, freezes
o: 3 frames
Student π_S
CNN + MLP
a_S
a_T w.p. β
β: 1→0
pose
$\mathcal{L} = \|a_S - a_T\|^2$
label
β
Simulator
512 parallel drones
+ domain rand.
s
Teacher π_T
(frozen)
a_T
executed action
(c) Deployment
no state estimator
no privileged information
FPV
camera
video
Student
π_S
sticks
Drone
(simulated, or real
with an FC in acro mode)